Computation for the Improved Design

***Cross-Sectional Area**

Hidden equation

1 1/4” Pipe:

Hidden equation

3/4” Pipe:

Hidden equation

3/4” Hose:

Hidden equation

*** Flow rate**

Hidden equation

Top Branch (1 nozzle):

Hidden equation

Left, Right, and front branches (each has 6 nozzles):

Hidden equation

***Velocity**

Hidden equation

1 1/4” Pipe:

Hidden equation

3/4” Pipe:

Hidden equation

3/4” Hose:

Hidden equation

Nozzle 3&4:

Hidden equation

Nozzle 2&5:

Hidden equation

Nozzle 1&6:

Hidden equation

***Reynolds Number**

Hidden equation

1 1/4” Pipe:

Hidden equation

3/4” Hose:

Hidden equation

3/4” Pipe, full flow rate:

Hidden equation

3/4” Pipe, nozzle 3:

Hidden equation

3/4” Pipe, nozzle 2:

Hidden equation

3/4” Pipe, nozzle 1:

Hidden equation

***Major Head loss (Darcy-Weisbach Equation)**

Hidden equation

Hidden equation, for 1 ¼’pipe, ¾’ pipe, and ¾’ hose

Hidden equation, friction factor

1 1/4” Pipe:

Hidden equation

3/4” Pipe:

Hidden equation

3/4” Hose:

Hidden equation

3/4” Pipe, nozzle 3:

Hidden equation

3/4” Pipe, nozzle 2:

Hidden equation

3/4” Pipe, nozzle 1:

Hidden equation

-Front Branch Head Loss

Length(L):

Hidden equation, for 1 ¼’ pipe

Hidden equation, for ¾’ hose

Hidden equation, for ¾’ pipe (full flow rate)

Hidden equation, for ¾’ pipe (nozzle3)

Hidden equation, for ¾’ pipe (nozzle2)

Hidden equation, for ¾’ pipe (nozzle1)

Hidden equation

Hidden equation

Hidden equation

Hidden equation

Hidden equation

Hidden equation

Hidden equation

Hidden equation

-Length and right Branches Head Loss

Length(L):

Hidden equation, for 1 ¼’ pipe

Hidden equation, for ¾’ hose

Hidden equation, for ¾’ pipe (full flow rate)

Hidden equation, for ¾’ pipe (nozzle3)

Hidden equation, for ¾’ pipe (nozzle2)

Hidden equation, for ¾’ pipe (nozzle1)

Hidden equation

Hidden equation

Hidden equation

Hidden equation

Hidden equation

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Hidden equation

Hidden equation

-Top Branch Head Loss

Length(L):

Hidden equation, for 1 ¼’ pipe

Hidden equation, for ¾’ hose

Hidden equation

Hidden equation

Hidden equation

Hidden equation

***Minor Head loss**

Hidden equation

-Left and Right Branches (Since Symmetrical)

- **Setup:** 1/8’ M to 1 ¼’ M reducer (Expanding) Hidden equation 1 ¼’ female threaded fitting Hidden equation 1 ¼’ Cross fitting (Branch flow) Hidden equation 1 ¼’ female threaded fitting Hidden equation
- 1 ¼’ M to ¾’ M adapter (Contracting) Hidden equation ¾’ solenoid valve Hidden equation
- ¾’ M to M Adapter Hidden equation ¾’ Bullhead tee Hidden equation
- ¾’ female threaded Tee (Straight flow) Hidden equation
- ¾’ female threaded tee (Straight flow) Hidden equation
- ¾’ female threaded elbow Hidden equation ¾’ M to ¼’ M adapter (Contracting)

Hidden equation, for 1 ¼’ size

Hidden equation, for ¾’ hose

Hidden equation, for ¾’ pipe

Hidden equation, for nozzle 3

Hidden equation, for nozzle 2

Hidden equation, for nozzle 1

Hidden equation

Hidden equation

Hidden equation

Hidden equation

Front Branch Setup

- **Setup:** 1/8’ M to 1 ¼’ reducer (Expanding Flow) Hidden equation 1 ¼’ Female threaded fitting Hidden equation 1 ¼’ Cross fitting (Straight flow) Hidden equation 1 ¼’ female threaded tee (Straight flow) Hidden equation 1 ¼’ female threaded fitting Hidden equation
- 1 ¼’ M to ¾’ M Adapter (Contracting) Hidden equation ¾’ Solenoid Valve Hidden equation ¾’ M to M adapter Hidden equation
- ¾’ M to M adapter Hidden equation ¾’ Bullhead tee Hidden equation
- ¾’ Female threaded tee (Straight flow) Hidden equation
- ¾’ Female threaded tee (Straight flow) Hidden equation
- ¾’ Female threaded elbow Hidden equation ¾’ M to ¼’ M adapter

Hidden equation, for 1` ¼’ pipe

Hidden equation, for ¾’ hose

Hidden equation, for ¾’ pipe

Hidden equation, for nozzle 3

Hidden equation, for nozzle 2

Hidden equation, for nozzle 1

Hidden equation

Hidden equation

Hidden equation

Hidden equation

-Top Branch

- **Setup:** 1/8 M to 1 1/4 M reducer (Expanding) Hidden equation 1 1/4 female threaded fitting Hidden equation 1 1/4 Cross fitting (Straight flow) Hidden equation 1 1/4 female threaded tee (branch flow) Hidden equation
- 1 1/4 M to 3/4’ M adapter (Contracting) Hidden equation 3/4’ solenoid valve Hidden equation 3/4’ M to M Adapter Hidden equation 3/4’ M to 1/4 M adapter (Contracting)

Hidden equation, for 1 ¼’ size

Hidden equation, for ¾’ hose

Hidden equation

Hidden equation

Hidden equation

Hidden equation

***Total head loss** Hidden equation

Branch	Major head loss	Minor Head loss	Total head loss
Left	0.1777076053 m	0.6446526127 m	0.822360218 m
Right	0.1777076053 m	0.6446526127 m	0.822360218 m
Front	0.1799413428 m	0.6396899163 m	0.8196312591 m
Top	0.1277037368 m	0.5774091181 m	0.7051128549 m

***Outlet pressure**

Hidden equation

-Left and right branches

Hidden equation

Hidden equation

-Front branch

Similar to left and right branches

-Top branch

Hidden equation

Hidden equation

***Hydraulic Power**

Hidden equation

Hidden equation

Hidden equation

Hidden equation

***Pump Input Power**

Hidden equation

Hidden equation, based on the computed hydraulic power and pump specification.

Hidden equation

Hidden equation

***Total Water Consumption, 30mins of operation**

Hidden equation

Hidden equation

Hidden equation

***Electrical Energy Consumption, 30mins of operation**

Hidden equation

Hidden equation

Hidden equation